Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing
Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 26-03-2026

Relational Inventory Control & Stock Tracking System

SDG: SDG No – SDG9- Industry, Innovation and Infrastructure

Presented By:

OBULAREDDY HARSHITHA– VTU25985

Reg.No.23UECS0409

Courses Handling Faculty:

Dr . HAFIZUR RAHMAN

INTRODUCTION

1
2
3
4
5
6
7
8
9

- Inventory management is an essential part of any business to track products, stock levels, and transactions efficiently.
- Traditional manual systems are time-consuming, error-prone, and difficult to maintain as business grows.
- This system helps store and organize data in tables such as products, suppliers, orders, and stock details.
- It allows real-time tracking of stock levels, reducing the risk of overstocking or stock shortages.
- The system improves accuracy by automating calculations and reducing human errors.
- This project develops a Relational Inventory Control & Stock Tracking System to manage inventory efficiently.
- The system uses a relational database to store data in structured tables, making it easy to access and manage information.
- It is scalable, meaning it can handle increasing data as the business grows.
- It can be used in various domains such as retail stores, warehouses, and e-commerce platforms.

PROBLEM STATEMENT

1

2

- Many businesses still use manual methods to manage inventory, which are slow and difficult to maintain.
- Manual record-keeping often leads to errors such as incorrect stock levels and missing data.
- There is no proper system to track real-time stock availability and movement.
- Businesses face problems like overstocking or running out of stock due to poor tracking.
- Data is not organized properly, making it hard to search and update information.
- Lack of automation increases workload and reduces overall efficiency.
- There is a need for a reliable and efficient system to manage inventory accurately.
- Managing large amounts of inventory data becomes complex without a proper system.
- Limited accessibility, as manual records cannot be accessed anytime or from anywhere.

3

4

5

6

7

8

9

OBJECTIVE(S)

1
2
3
4
5
6
7
8
9

- To develop a system that helps businesses manage inventory efficiently, reducing time and effort compared to manual methods.
- To organize and store all inventory-related data using a relational database, so that information is structured and easy to access.
- To minimize errors in stock management by automating routine operations such as adding, updating, and deleting inventory records.
- To generate reports and analytics that assist in business decision-making and understanding stock trends.

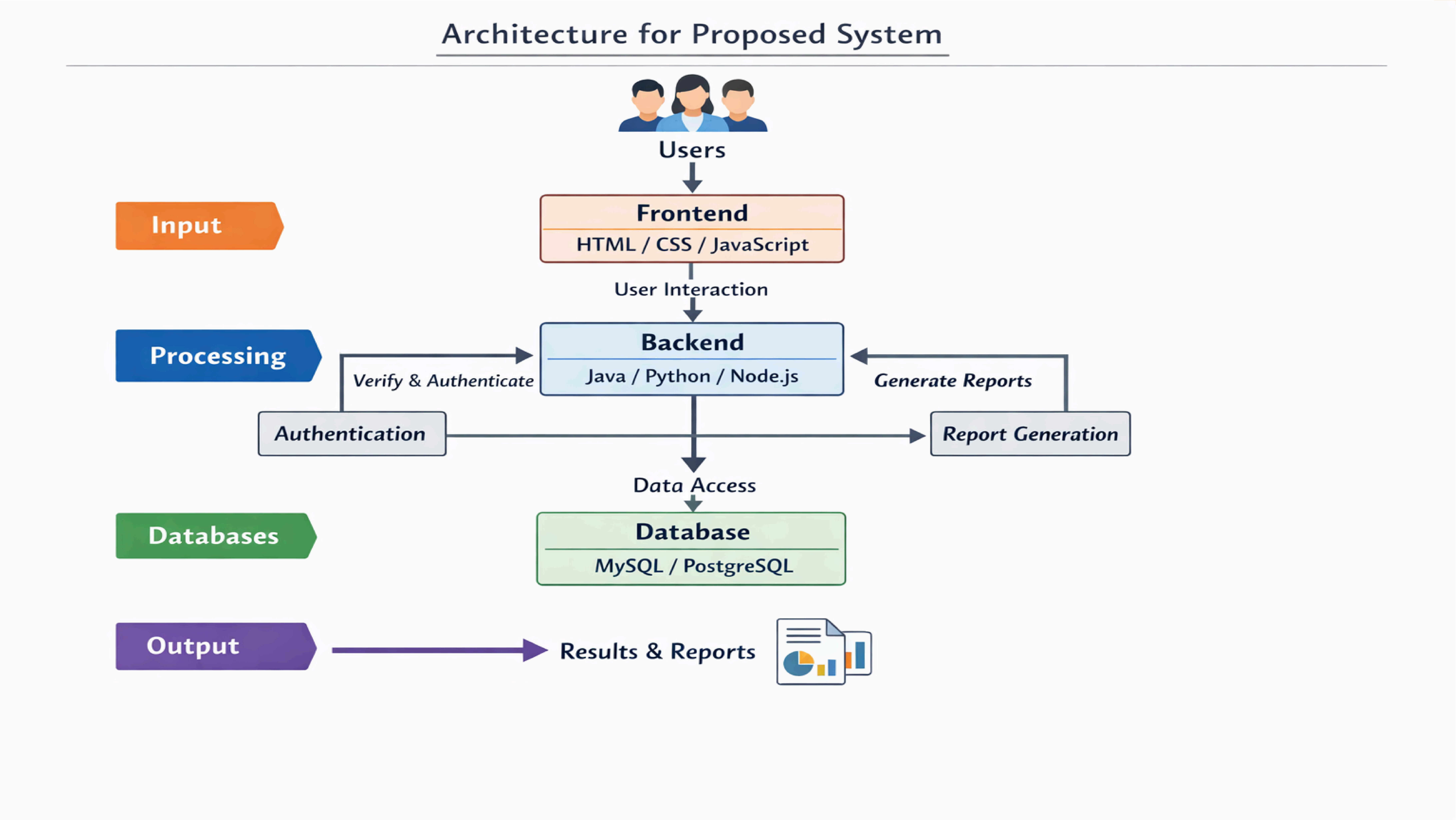
PROPOSED SYSTEM

1
2
3
4
5
6
7
8
9

- The proposed system manages inventory and stock digitally, replacing slow and error-prone manual methods.
- Users can add new stock, update existing records, delete obsolete items, and view inventory details, making stock management efficient and simple.
- Stock levels are updated in real-time whenever items are added, sold, or removed, helping businesses prevent shortages or overstocking.
- All product, supplier, and transaction details are stored in a relational database, ensuring data is structured, secure, and easy to retrieve when needed.
- The system has a login module to allow only authorized users to access it.

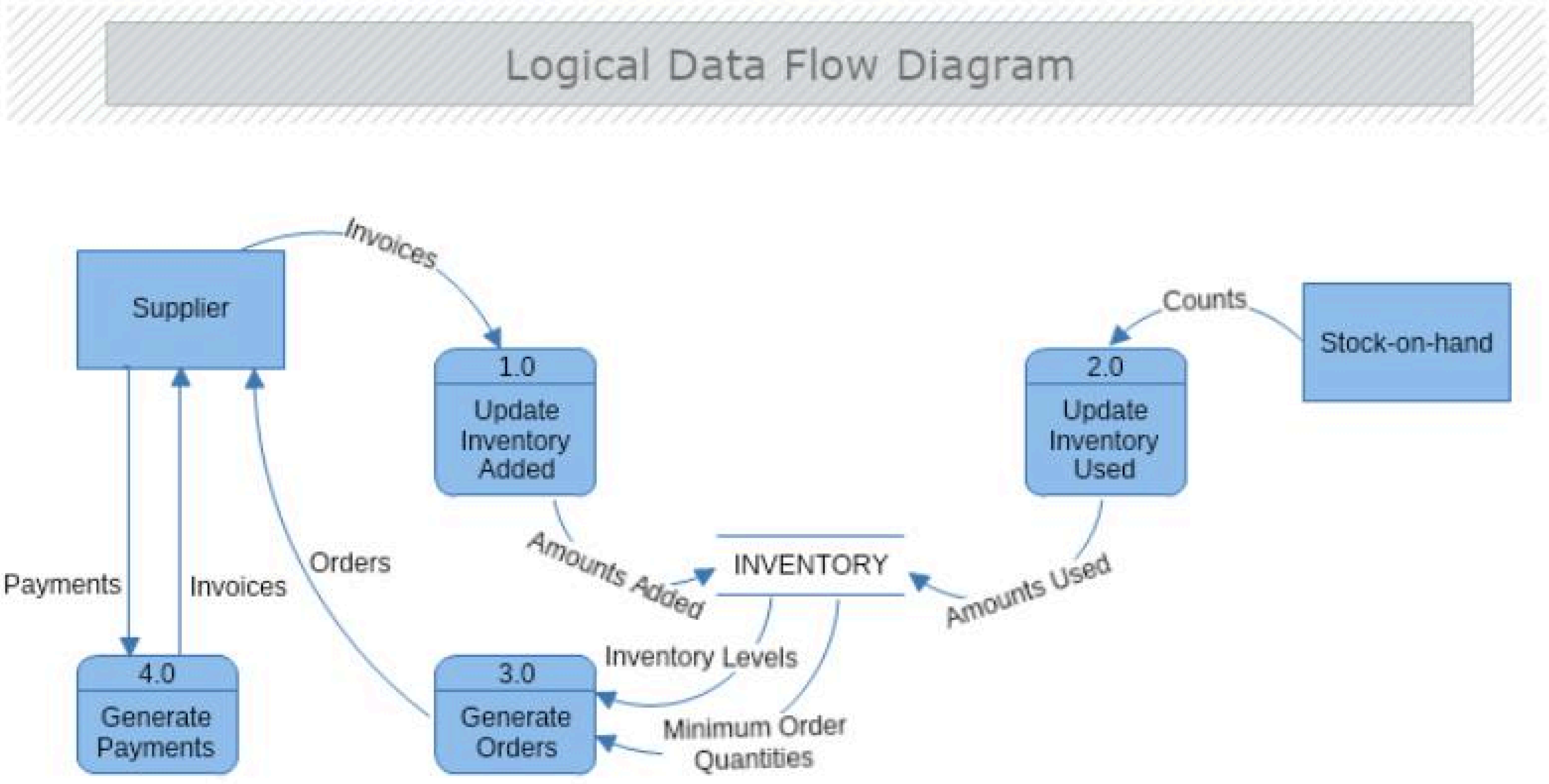
PROPOSED SYSTEM ARCHITECTURE

1
2
3
4
5
6
7
8
9



WORKFLOW

1
2
3
4
5
6
7
8
9



TECHNOLOGIES USED

- **Frontend:**
- HTML
- CSS
- JavaScript
- **Backend:**
- Node.js
- Express.js
- **Database:**
- MySQL
- **Tools:**
- Vs Code
- **Hardware/Software Requirements:**
- Intel i3 or above
- Windows ,OS
- Web Browser(Chrome/Edge for http://localhost:5000/signup.html)

MODULE DESCRIPTION

Module 1: Product Management Model

- The Product Management Model is responsible for handling all product-related information in the system. It allows users to add new products by entering details such as product name and quantity.
- Enables deletion of products.

Module 2: Inventory Control Model

- This model also helps in identifying low stock situations, which allows users to take necessary actions such as restocking.
- Helps monitor current inventory status.

Module 3: Database Management

- This module connects the application to the MySQL database.
- It performs data insertion, retrieval, updating, and deletion operations.

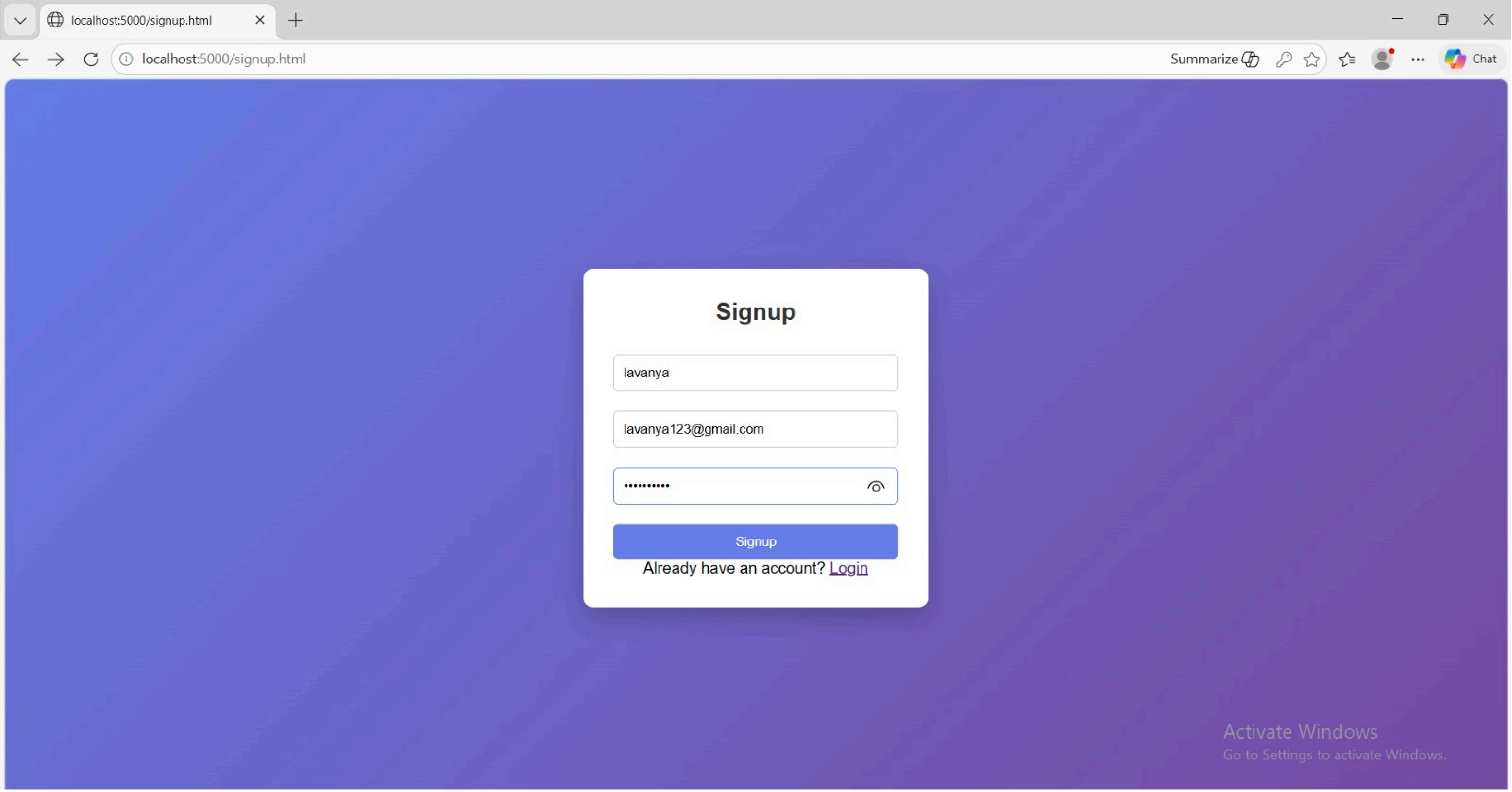
Module 4: User Interface Model

- This model ensures that the system is easy to use, accessible, and provides a smooth user experience.
- Displays data in tabular format.

IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8
9

- Allows new users to create an account by entering required details such as name, email, and password with proper validation.
- The entered information is securely stored in the database, enabling users to log in and access the system features.



IMPLEMENTATION SCREENSHOTS

1

2

3

4

5

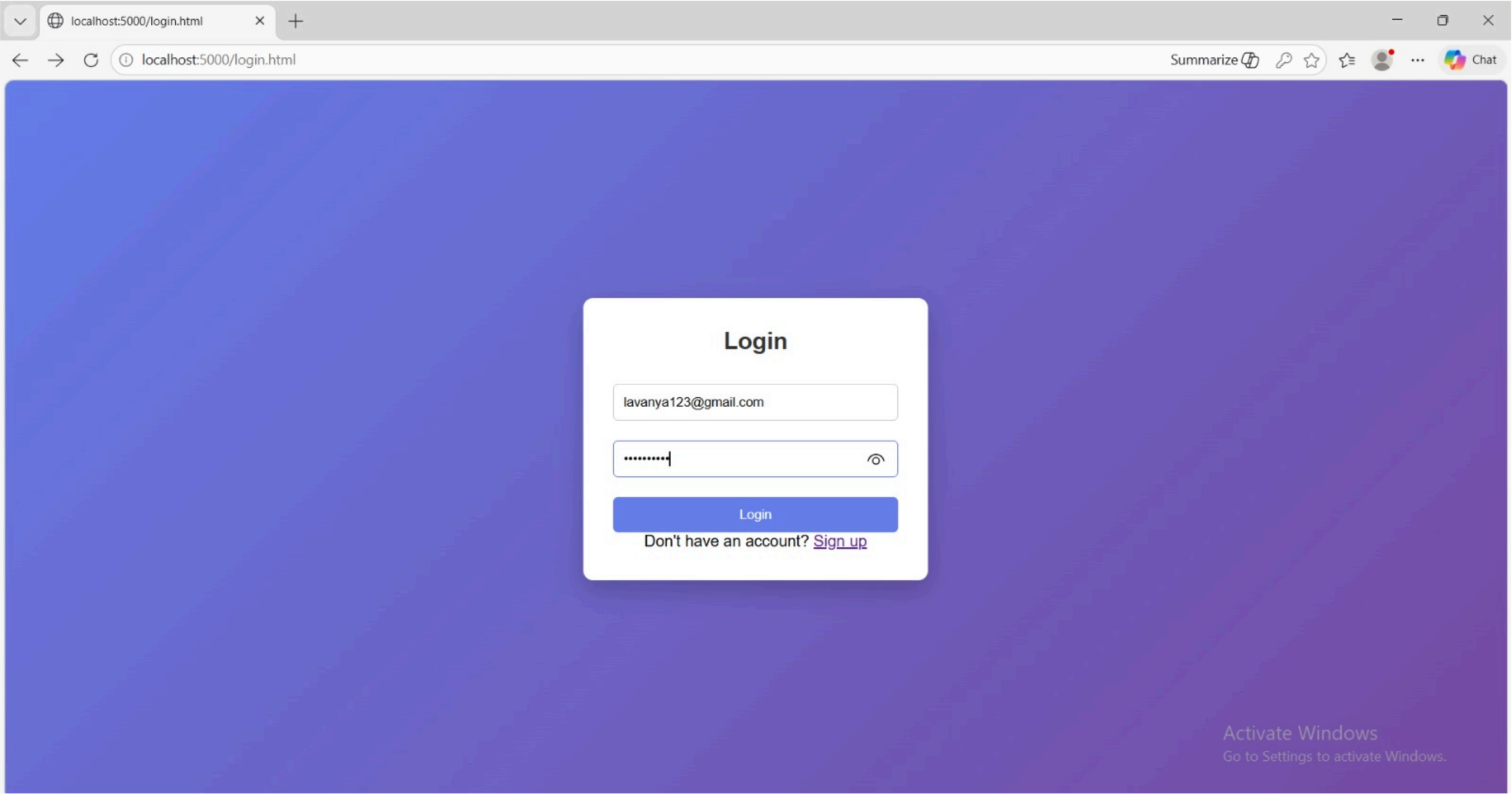
6

7

8(A)

9

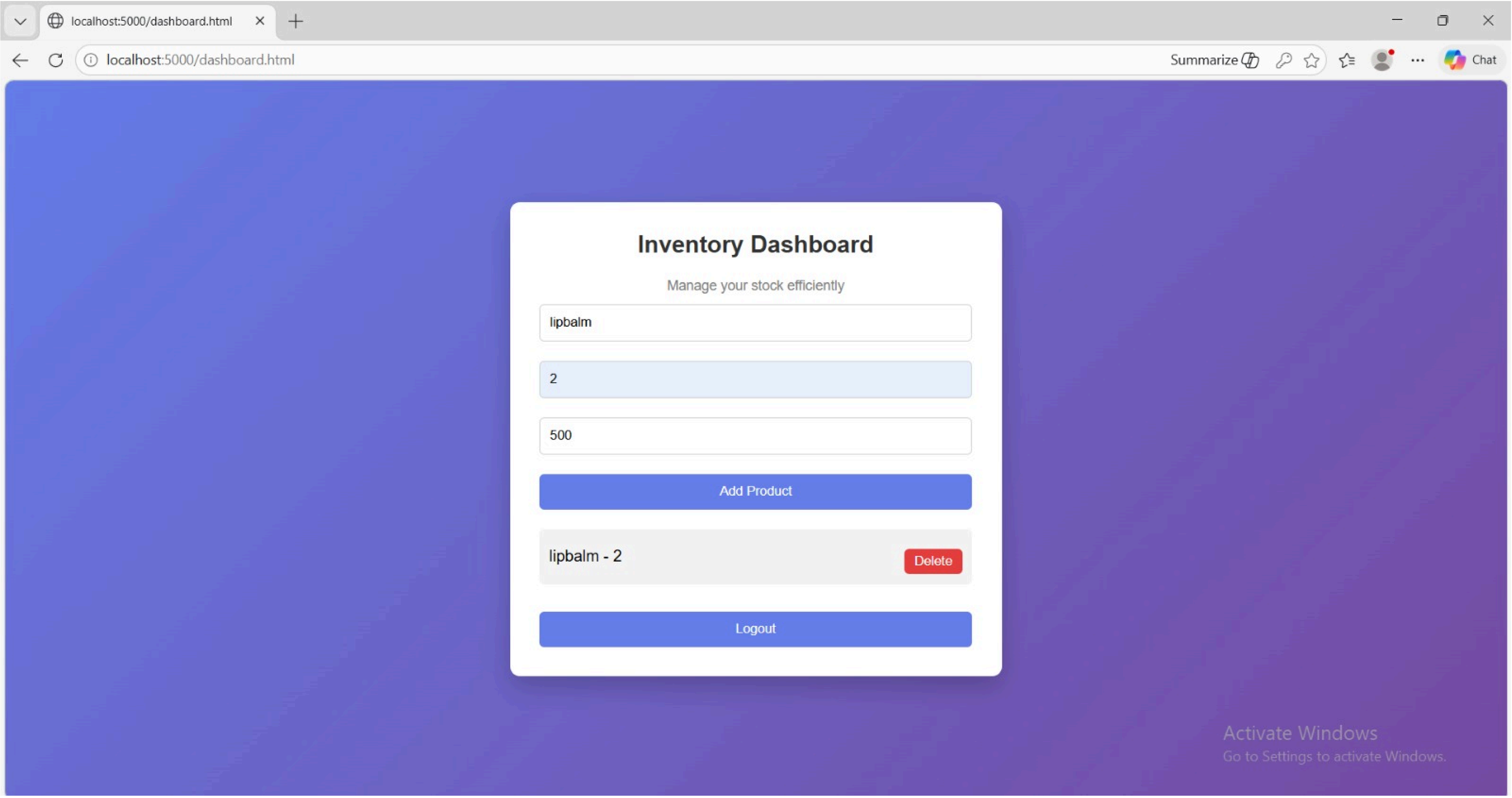
- Allows registered users to enter their email and password to securely access their account within the system.
- The system verifies the credentials with the database and grants access to the dashboard upon successful authentication.



IMPLEMENTATION SCREENSHOTS

1
2
3
4
5
6
7
8(B)
9

- Admin can add new products with name and quantity into the system.
- Ensures proper stock entry and maintains accurate inventory records.



ADVANTAGES & LIMITATIONS

1
2
3
4
5
6
7
8
9

Advantages:

- The system provides real-time updates of inventory, helping to avoid situations like overstocking or running out of products.
- It supports scalability, meaning the system can handle increasing inventory and data as the business grows.
- Helps prevent overstocking or running out of products.
- Ensures secure access with login for authorized users only.

Limitations:

- The system requires a computer and internet connection to operate.
- Large amounts of data may slow down the system if not optimized properly.
- Security depends on proper authentication; weak passwords can be a risk.

Thank you